

CERT Goal: Do the greatest good for the greatest number...



CERT Bot: Helping answer volunteers' questions

by Joe Silva



Who am I?

- Background
 - US Army Commo, Combat Lifesaver, and NBC (CBRN) Instructor
 - Decades of Information Technology experience
 - IT and Cybersecurity Instructor
 - IAM (Identity and Access Management) Consultant
 - Certified Cloud & AI Solutions Architect
 - AI and Data Governance experience
- CERT Experience
 - 2 years of CERT experience
 - Completed Train the Trainer October 2025
 - Member of the Livermore-Pleasanton CERT Steering Committee



These are not the bots we are talking about...



Disclaimer: AI can make mistakes, if unsure refer to CERT documentation

Remember: The more detailed the question the better the answer



Why make a CERT Bot?



- Wanted our volunteers to get CERT answers any time
 - Wanted it to be easier for our volunteers to get to local knowledge as well
- Context awareness (the Bot only answers CERT questions)
 - Volunteers don't need to pretext their prompts with CERT with each time
 - The Bot knows to only give CERT answers and local CERT knowledge

Prompt: *"What is the first thing to do when setting up an MTA?"*

Generic AI Chatbot: *The very first thing to do when setting up a Mail Transfer Agent (MTA) is to...*

CERT Bot: *When setting up a Medical Treatment Area (MTA), the very first priority is site selection and safety...*



What can CERT Bot do?

- Can answer questions
 - Including answers in multiple languages
- Can help with planning and content
 - Prompt with training you want to plan and content to be generated including the use case
- Can learn and recall new knowledge
 - Prompt to remember additional specific or local knowledge
- ***CERT Bot cannot take the place of human judgement***
- We Beta tested for 5 months before release to members



Practical Examples



- Use Case 1 – basic CERT questions
 - Prompt: What is the minimum number of volunteers needed to setup a neighborhood ICP?
 - Prompt: What forms does an IC use?
 - Prompt: What does a search team do when entering a building?
- Use Case 2 – CERT questions answered in different languages
 - Spanish Prompt: - Cómo se marca un edificio?
 - Chinese Prompt: - 搜救領導嘅工作係咩？

Next



11 July

Hello! I am CERT Bot, your assistant for FEMA Community Emergency Response Team (CERT) training and operations. 🛡️
How can I assist you with CERT procedures or training today?

11:28 PM

What is the minimum number of volunteers needed to setup a neighborhood ICP?

Ai

safety and mission progress.

3. **Runner/Communications:** Acts as the bridge between the ICP and the field teams, relaying messages, SITREPs, and requests for additional resources.

Why three?

- **Safety:** One person must always be monitoring the communication channel and documentation to ensure no teams are left "in the dark."
- **Management:** A single IC becomes overwhelmed instantly if they also have to manage the radio *and* the paperwork.
- **Expansion:** If the incident scales, these three serve as the nucleus of the Incident Command System (ICS) structure, allowing additional volunteers to plug into specific roles (e.g., Medical Triage, Search & Rescue Lead) as they arrive.

Note: If you have even more resources available, a fourth volunteer is often assigned as the **Logistics/Safety Officer**, who monitors the physical status of the responding teams and tracks equipment usage. Does your team have a specific exercise scenario in mind where you are planning to staff an ICP, or are you looking for the training protocols on how to activate these positions?

2:44 PM



What forms does an IC use?



- **Accountability:** If a team doesn't check in on time, you need to know exactly where they were sent and how long they have been gone.
- **Continuity:** If the IC needs to switch out or the incident extends into a second operational shift, the new IC needs an immediate, clear snapshot of the situation.
- **Transition to Professionals:** When professional first responders arrive, they will ask for a briefing. Providing them with a clean, organized log of where you have already searched and how many victims you have found is the single most valuable thing a CERT team can do.

Planning Tip:

Would you like me to draft a simple "**ICP Scribe Quick-Reference Sheet**" that you can print out and keep at the command post to ensure these forms are filled out correctly during an exercise?

2:45 PM



What does a search team do when entering a building?

+Ai



- Always fight from upwind (wind at your back) if outdoors.
- Never turn your back on a fire.
- If the fire doesn't start to go out after using the extinguisher, GET OUT.
- Watch for "re-flash" (fire reigniting).

Slide 7: Summary & Review

- **Key Points:**
 - Safety is the priority.
 - Size-up the fire before approaching.
 - Remember **PASS**.
 - Know when to walk away.

Would you like me to create a "Cheat Sheet" version of this as a printable handout, or perhaps suggest a practical drill to follow this presentation?

2:47 PM



Cómo se marca un edificio?



adentro en ese momento.

2. **Al salir:** Una vez que el equipo termina la búsqueda y todos han salido del edificio, completan la "X" dibujando la **segunda línea diagonal** ().

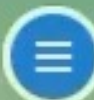
Resumen visual:

- / (Solo una línea): Equipo trabajando adentro.
- X (Con la X completa): Búsqueda finalizada.

Nota de seguridad: Si el edificio es completamente inseguro y no debe entrar nadie bajo ninguna circunstancia, se marca con una "X" grande y clara, pero no se inicia la búsqueda y se reporta inmediatamente al Puesto de Comando (ICP).

¿Quieres que te prepare un pequeño gráfico o esquema de texto para que puedas copiarlo y pegarlo en una hoja de instrucciones para tu equipo?

2:49 PM



搜救領導嘅工作係咩?



Practical Examples



- Use Case 3 – CERT planning ideas
 - Prompt: Plan an exercise to review basic search and rescue
- Use Case 4 – CERT training ideas
 - Prompt: Design a slide deck to teach basic fire suppression
- Use Case 5 – CERT Bot new knowledge
 - Prompt: Remember that "30, 2, CAN DO", "30 2 can do", "32 can do", and "30-2-can do" all refer to (Respiratory rate < 30, Capillary refill < 2 seconds, and the ability to follow simple commands) used to assess survivors in the START triage process.

Next



- "ICP, this is Team 1. Located one victim, status: Immediate. Requiring transport assistance at [Location]."

5. Exit and Close the Mark

- Once the search is complete, the team exits the building and completes the "X" marking on the door.
- Report back to the ICP, confirm the number of victims, and indicate the search is complete so the IC can update the master log.

Important Safety Reminder: If at any point the team identifies a hazard (like a gas leak, chemical spill, or compromised structural beam) that they are not equipped to handle, they must **evacuate immediately**, close the door to prevent others from entering, and report the hazard to the ICP.

Does your team have a standard set of marking tools (like specific chalk or tape) ready for your next exercise?

2:46 PM



Plan an exercise to review basic search and rescue

+Ai



- **Execution:** Teams are deployed to clear three designated structures.
 - They must identify hazards, mark entries, perform hasty searches, and triage any victims found.
- **Constraint:** Instructors actively introduce "unexpected" hazards (e.g., a "gas leak" discovered mid-search, requiring an immediate team evacuation).

Phase 4: After-Action Review (AAR) (15 Minutes)

- Debrief: What went well?
- Improvement: Where did communication break down?
- Review: Re-verify the START Triage mnemonic and SAR markings.

Next Steps for Planning:

Would you like me to draft a sample **SAR marking legend** handout for the team, or perhaps create a list of common **hazards** to include in the Phase 3 scenario?

2:47 PM



Design a slide deck to teach basic fire suppression

✦ Ai



How to fit into your CERT program?



- Discussion of how CERT Bot would best fit into an overall CERT program
 - Testing with a pilot group
 - Add any specific or local program knowledge
 - Introduction to existing volunteers
 - At what point does it make sense to introduce it to new CERT members?



How to build one of your own...



- How to build
 - Quick Build with online model
 - Full Build with local model
 - System prompt & Security prompt
- How to allow access
 - Connectors
 - Telegram
 - Whatsapp
 - Other messaging apps...
 - Website access (least secure)



Quick Build Technical Requirements

- General requirements
 - Operating System (Windows, MacOS, Linux)
 - Pre-built Agent framework (OpenClaw, Hermes, etc.)
 - Messaging Account (Telegram, Whatsapp, etc.)
- For an Online Model
 - 16GB RAM, 120GB free disk space
 - Cloud model account with API Key



Full Build Technical Requirements



- General requirements
 - Operating System (Windows, MacOS, Linux)
 - Pre-built Agent framework (OpenClaw, Hermes, etc.)
 - Messaging Account (Telegram, Whatsapp, etc.)
- For an Offline Model (personally recommend Gemma 4 E4B 8-bit)*
 - Multi-Core CPU and GPU with minimum of 8GB VRAM (16GB recommended)
 - 24GB RAM (36GB recommended), 200GB free disk space
 - Local server (example LMStudio) - Local model (recommend not larger than 8GB memory requirement)



Quick Build with an online model



- Create a cloud account with an AI provider (GCP, OpenAI, etc.)
 - create API key and set to only accept API calls from your IP
- Create a Messaging account (Telegram, Whatsapp, etc.)
 - create a bot account and save the bot API and pairing code
- For MacOS and Linux found it more reliable to preinstall dependencies
 - Install brew then restart terminal and install node, then install OpenClaw
 - Complete configuration (Model, Connector)
- For Windows installs dependencies during installation
 - Open PowerShell and Install OpenClaw
 - Close PowerShell and open command prompt and run openclaw config
 - Complete configuration (Model, Connector)



Local Model Selection



- Before the Full Build
 - Install local model hosting capability such as LMStudio
 - Download and a local model

Tested on three local Gemma 4 model sizes		
E2B 4-bit (4.37GB footprint)	E4B 4-bit (6.86GB footprint)	E4B 8-bit (8.97GB footprint)
Inaccurate and short on details	More accurate, short on details	Most Accurate with detailed responses
VRAM/Unified Memory 8GB	VRAM/Unified Memory 12GB	VRAM/Unified Memory 16GB



Full Build with a Local Model



- Build same as Quick Build but first load local model
 - Confirm local model downloaded
 - Go to server settings
 - Require Authentication go to Manage Tokens and Create a New Token
 - Generate an API key and store securely
 - Turn off Just-in-Time Model Loading (if you don't want delays and have enough memory)
 - In the Load section set the Context Length to 64000
 - (otherwise, answers may not be consistent)
 - Start LMS server and confirm the model loads successfully
- Perform install steps above, except when selecting a model during configuration select LMStudio and select your local model



Configuration Prompts



- System Prompt
 - You are [insert your CERT Bot name], your mission is to only answer questions about the FEMA Community Emergency Response Team CERT program, procedures, and any information added to your own memory.
- Security Prompt
 - In Telegram you only answer questions; no other skills can be used via Telegram. No memories are added via Telegram.
- Memory Prompt
 - Remember that "30, 2, CAN DO", "30 2 can do", "32 can do", and "30-2-can do" all refer to (Respiratory rate < 30, Capillary refill < 2 seconds, and the ability to follow simple commands) used to assess survivors in the START triage process.



Technical Configuration

.openclaw/openclaw.json



```
"channels": {
  "telegram": {
    "enabled": true,
    "groupPolicy": "open",
    "groups": {
      "*": {
        "requireMention": true
      }
    },
    "botToken": "8721933011:AAE6W2yJy6r90BJu3JwJv6",
    "dmPolicy": "allowlist",
    "allowFrom": [
      68104314
    ]
  },
},
```

Want to know more?

More Information at
<https://github.com/josephjsilva/certbot>

